-인체조직 특성의 초탄성 TiNi 의료용 소재 국산화기업-

㈜TiNiKo(구, ㈜강앤박메디컬) 社 제품 및 회사 소개

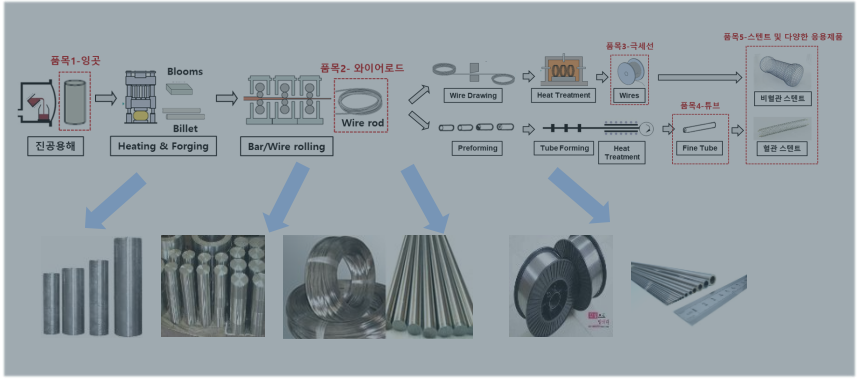
주식회사 티니코
강지훈 대표이사

All about that, TiNi Shape Memory Alloys, TiNiAll
TiNi-based Superelastic Bio-materials of KOREA, TiNiKo

tiniko

2024’ 10월

- 목 차 –



1. 생체의료용 초탄성 니티놀 소재
(=Nitinol, =TiNi superelastic biomaterial)

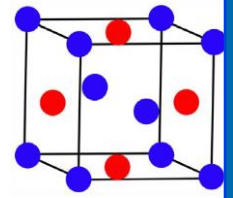
2. 기술 및 제품

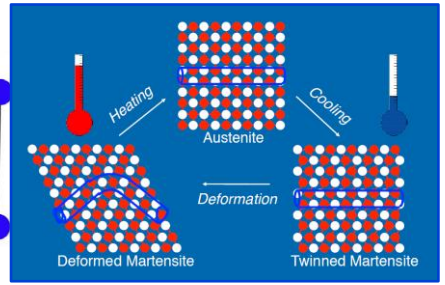
3. 판매 제품

Materials
BioMed
tiniko

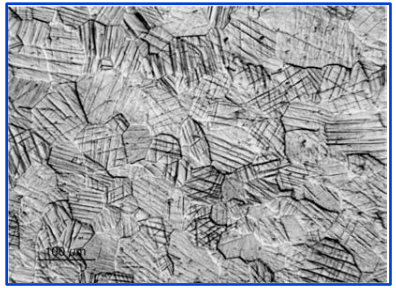
생체의료용 초탄성 Ti 합금 – 니티놀 (Nitinol, TiNi Shape Memory and Superelastic Biomaterials)

온도나 응력 조건 변화에 의해 변형부분이 원래의 모양으로 복원되는 특성을 가진 소재

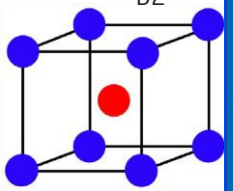


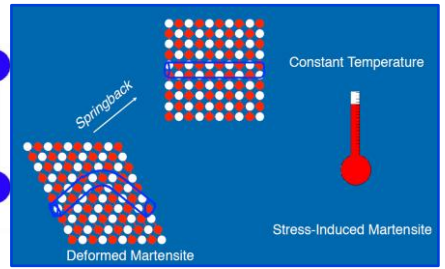


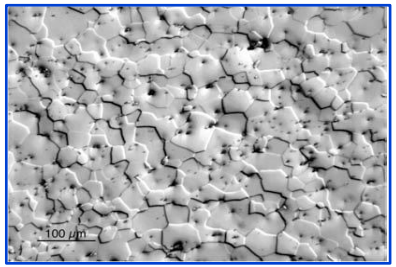
(격자구조 변화)

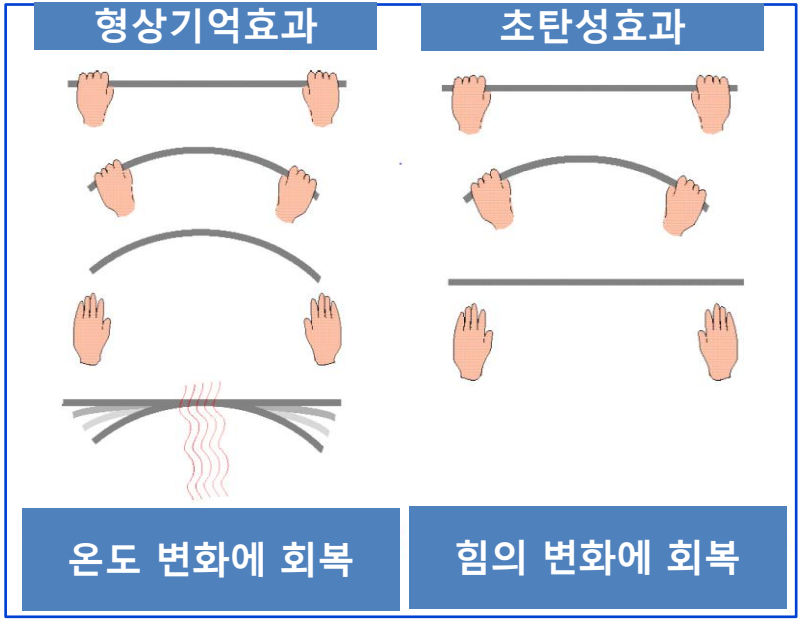


(미세조직 변화)



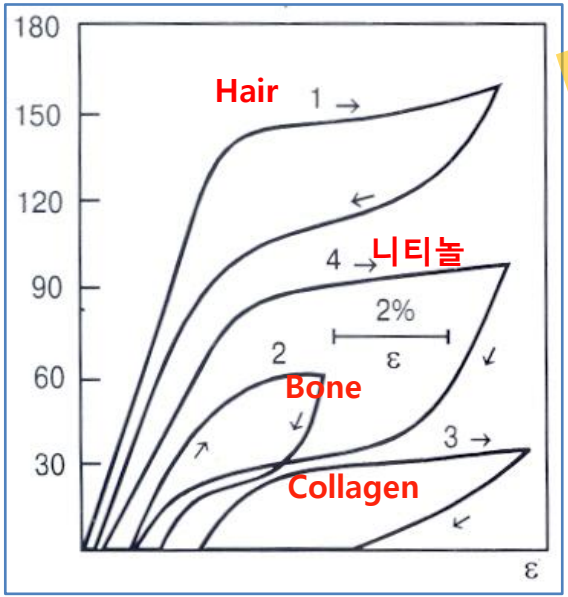






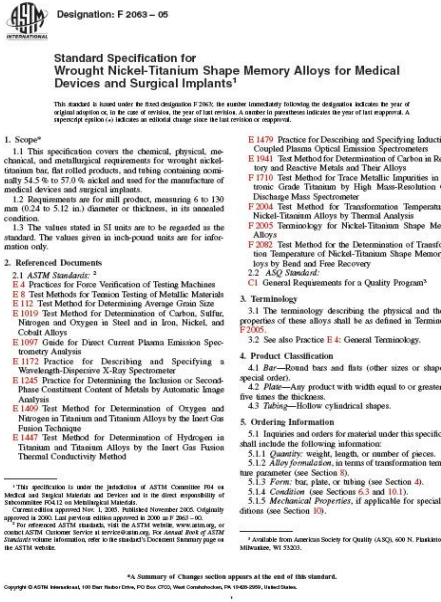
생체의료용 초탄성 Ti 합금 – 니티놀 (Nitinol, TiNi Shape Memory and Superelastic Biomaterials)

생체화학적, 생체역학적 적합성이 가장 우수한 ASTM F2063 규격의 생체이식용 의료소재



생체역학적특성

Biomechanical Compatibility
Dynamic Interference
Hysteresis / Biased Stiffness
Elastic Deployment
Fatigue Resistance
피로 특성
Thermal Deployment
Biocompatibility
Kink Resistance
MR Compatibility
MRI호환성
Constancy of Stress



Designation: F 2063 – 05

Standard Specification for Wrought Nickel-Titanium Shape Memory Alloys for Medical Devices and Surgical Implants[1]

This standard is issued under the fixed designation F 2063; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

1. Scope*

1.1 This specification covers the chemical, physical, mechanical, and metallurgical requirements for wrought nickel-titanium bar, flat rolled products, and tubing containing nominally 54.5 % to 57.0 % nickel and used for the manufacture of medical devices and surgical implants.

1.2 Requirements are for mill product, measuring 6 to 130 mm (0.24 to 5.12 in.) diameter or thickness, in its annealed condition.

1.3 The values stated in SI units are to be regarded as the standard. The values given in inch-pound units are for information only.

2. Referenced Documents

2.1 *ASTM Standards:* [2]

E 4 Practices for Force Verification of Testing Machines
E 8 Test Methods for Tension Testing of Metallic Materials
E 112 Test Method for Determining Average Grain Size
E 1019 Test Method for Determination of Carbon, Sulfur, Nitrogen and Oxygen in Steel and in Iron, Nickel, and Cobalt Alloys
E 1097 Guide for Direct Current Plasma Emission Spectrometry Analysis
E 1172 Practice for Describing and Specifying a Wavelength-Dispersive X-Ray Spectrometer
E 1245 Practice for Determining the Inclusion or Second-Phase Constituent Content of Metals by Automatic Image Analysis
E 1409 Test Method for Determination of Oxygen and Nitrogen in Titanium and Titanium Alloys by the Inert Gas Fusion Technique
E 1447 Test Method for Determination of Hydrogen in Titanium and Titanium Alloys by the Inert Gas Fusion Thermal Conductivity Method
E 1479 Practice for Describing and Specifying Inductively-Coupled Plasma Optical Emission Spectrometers
E 1941 Test Method for Determination of Carbon in Refractory and Reactive Metals and Their Alloys
F 1710 Test Method for Trace Metallic Impurities in Electronic Grade Titanium by High Mass-Resolution Glow Discharge Mass Spectrometer
F 2004 Test Method for Transformation Temperature of Nickel-Titanium Alloys by Thermal Analysis
F 2005 Terminology for Nickel-Titanium Shape Memory Alloys
F 2082 Test Method for the Determination of Transformation Temperature of Nickel-Titanium Shape Memory Alloys by Bend and Free Recovery

2.2 *ASQ Standard:*

C1 General Requirements for a Quality Program[3]

3. Terminology

3.1 The terminology describing the physical and thermal properties of these alloys shall be as defined in Terminology F2005.

3.2 See also Practice E 4: General Terminology.

4. Product Classification

4.1 *Bar*—Round bars and flats (other sizes or shapes by special order).

4.2 *Plate*—Any product with width equal to or greater than five times the thickness.

4.3 *Tubing*—Hollow cylindrical shapes.

5. Ordering Information

5.1 Inquiries and orders for material under this specification shall include the following information:

5.1.1 *Quantity*: weight, length, or number of pieces.

5.1.2 *Alloy formulation*, in terms of transformation temperature parameter (see Section 8).

5.1.3 *Form*: bar, plate, or tubing (see Section 4).

5.1.4 *Condition* (see Sections 6.3 and 10.1).

5.1.5 *Mechanical Properties*, if applicable for special conditions (see Section 10).

[1] This specification is under the jurisdiction of ASTM Committee F04 on Medical and Surgical Materials and Devices and is the direct responsibility of Subcommittee F04.12 on Metallurgical Materials.
Current edition approved Nov. 1, 2005. Published November 2005. Originally approved in 2000. Last previous edition approved in 2000 as F 2063 – 00.

[2] For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

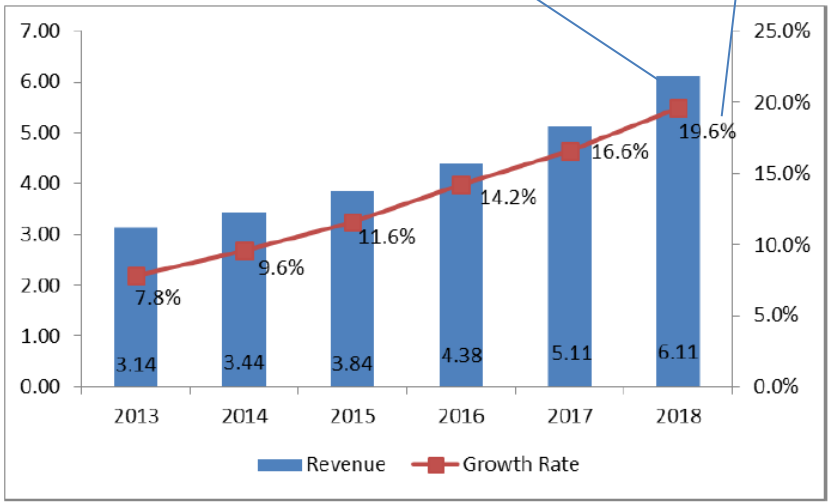
[3] Available from American Society for Quality (ASQ), 600 N. Plankinton Ave., Milwaukee, WI 53203.

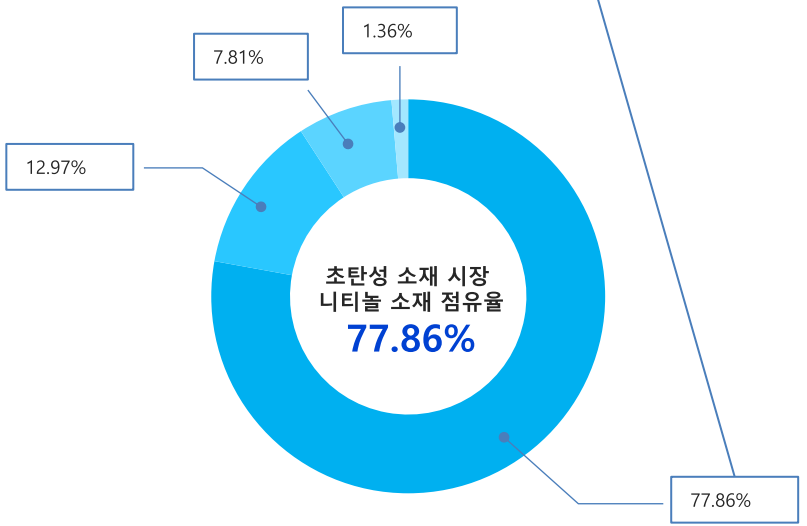
*A Summary of Changes section appears at the end of this standard.

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해외 시장 현황

연간 6조원 세계 시장규모, 19.6%의 성장율, 초탄성 소재 시장 77.86%의 시장 지배력

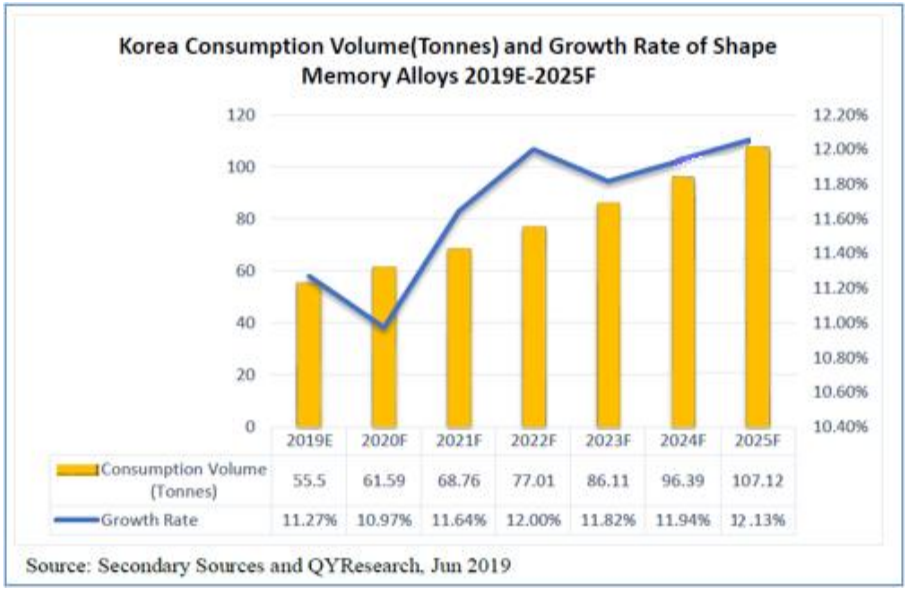




Source: TechNavio Analysis

국내 시장 현황

국내 산업분야에서의 연간 107톤 규모의 사용량 증가와 중견기업 수준에서의 수요 니즈 증가

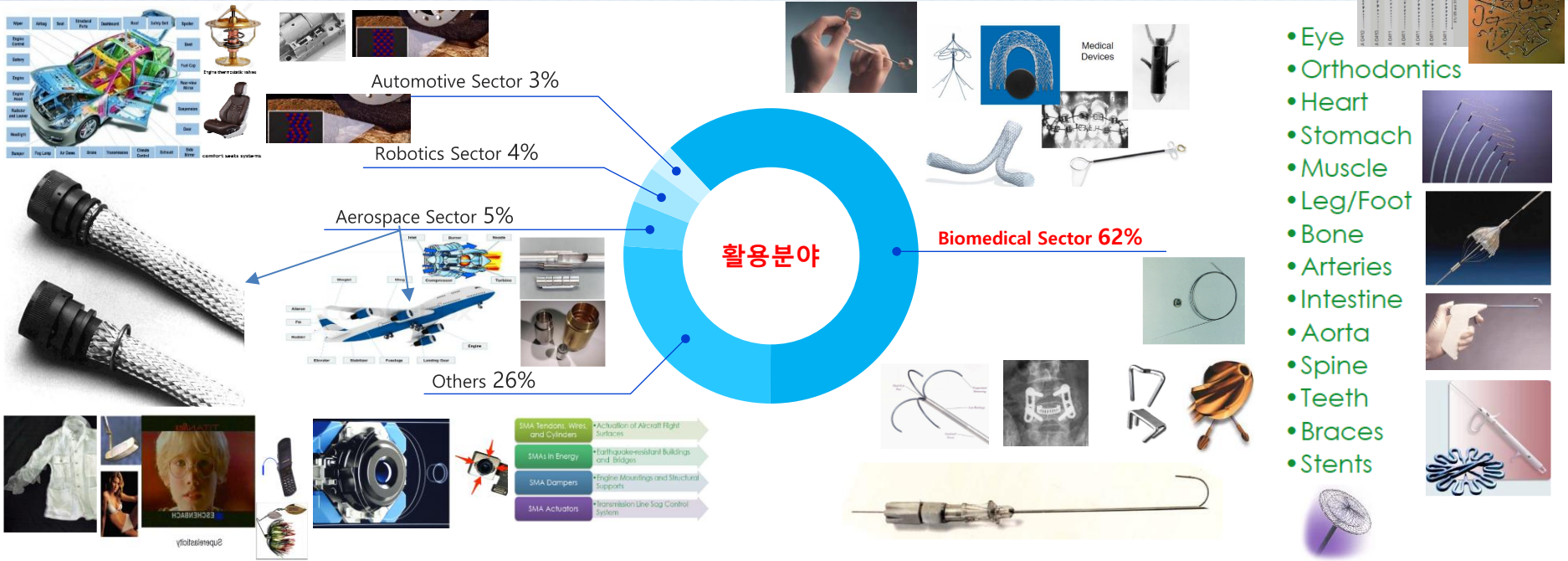


2025년 기준, 약 107톤의 예측수요
60%를 차지하는 의료산업 부분에 자립화를 가정
제작기간, 품목 회전율, 등을 고려 최소 필요 생산중량 추산

약 ,12톤 규모 (예측 수요량 전체의 12% 수준)로 추산됨.
이는 (1,000kg/월), (50-100kg/일) 생산중량 수준.
국내 수입업체, 가공업체, 등에서의 공급 희망 수렴 중량은 최소 30kg 이상급 니티놀 봉재(bar) 및 가공재(semi-finished products) 정도임..

국산화 필요성 1

우주, 항공, 국방, 자동차, 건설, 에너지, 산업안전, 의류산업, 로봇, 의료, 등 전방위 산업분야 활용



국산화 필요성 2

해외 의존도 100%인 소재로 특히, 대일 의존도 탈피가 절실한 전략적 핵심소재

(대일 무역 리스트에 적시된 Ti 소재 품목)

(대형 니티놀 원소재, 와이어, 등)



국내가공
극세선, 튜브



국내가공
스텐트 외 제품

현재 공급 구조
(일본, 등 해외 의존)

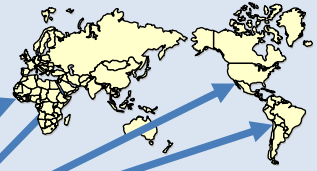
기술독립에 의한 공급구조 변화

국산 대형 니티놀 원소재
(㈜강앤박메디컬, 재료연구소)





국산 극세선, 튜브
(고려특수선재㈜, 삼화스틸㈜)

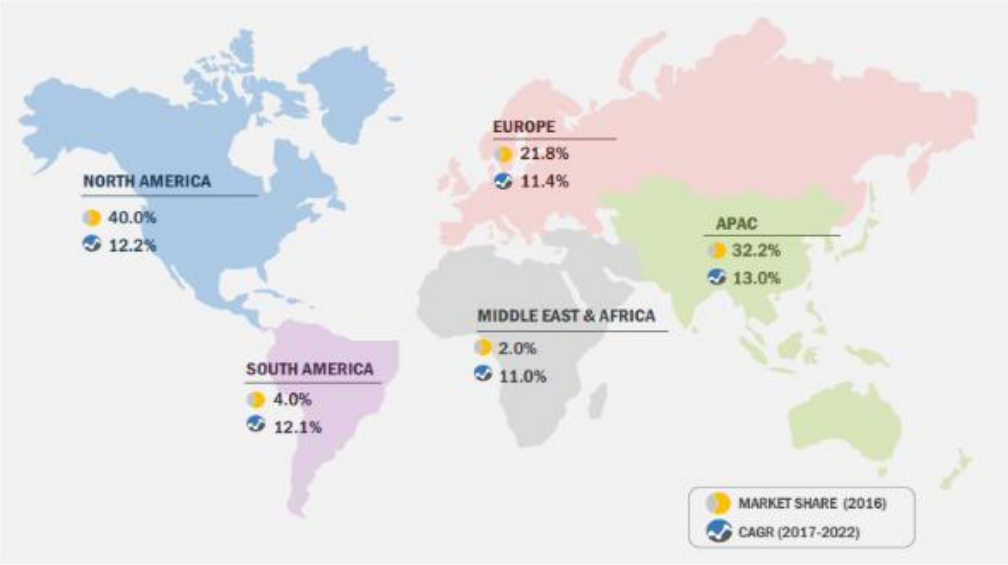


국산 스텐트 외 제품
(시지바이오, 전 참여기업)

당 사업 창출 공급 구조를 통한 자립화
(국산제품 국내·외 시장 진출)

국산화 필요성 3

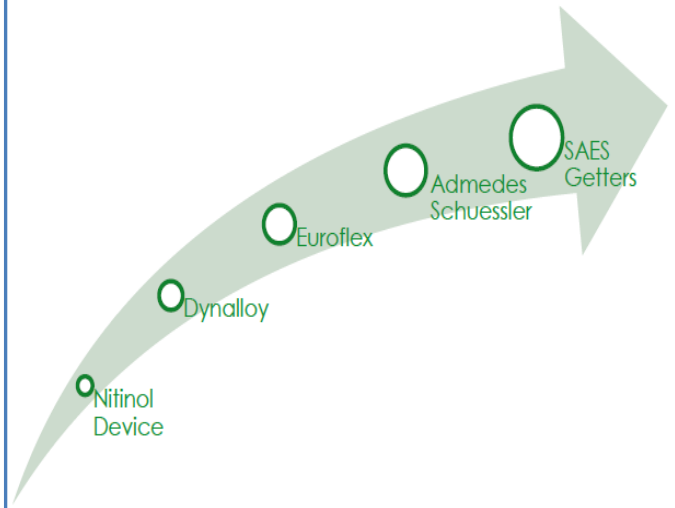
- 미국, 독일, 일본, 중국, 등 각국 대표 회사가 있는데, 한국은 ?
- 세계 점유율에서 일본, 중국이 각각 5.6%, 4.5% 차지하고 있는데, 한국은 ?
- 대륙별 시장점유율이 아시아가 제일 성장하고 있는데, 한국은 ?



Note: The market share and CAGR are provided in terms of value.

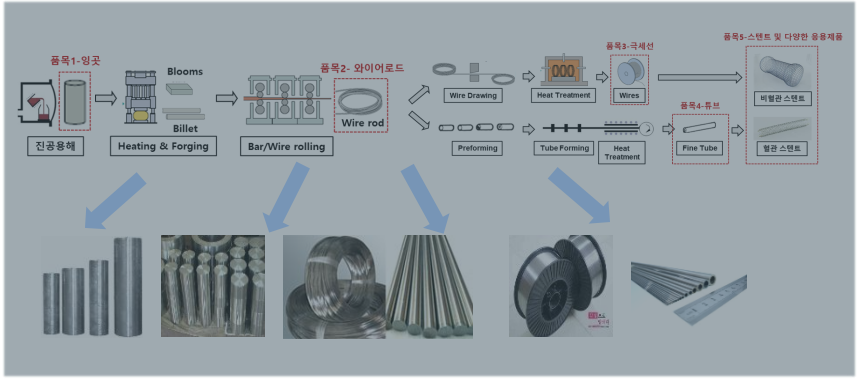
Source: Expert Interviews, Related Associations/Institutes, and MarketsandMarkets Analysis

세계랭킹



Source: TechNavio Analysis

- 목 차 –



1. 생체의료용 초탄성 니티놀 소재
(=Nitinol, =TiNi superelastic biomaterial)

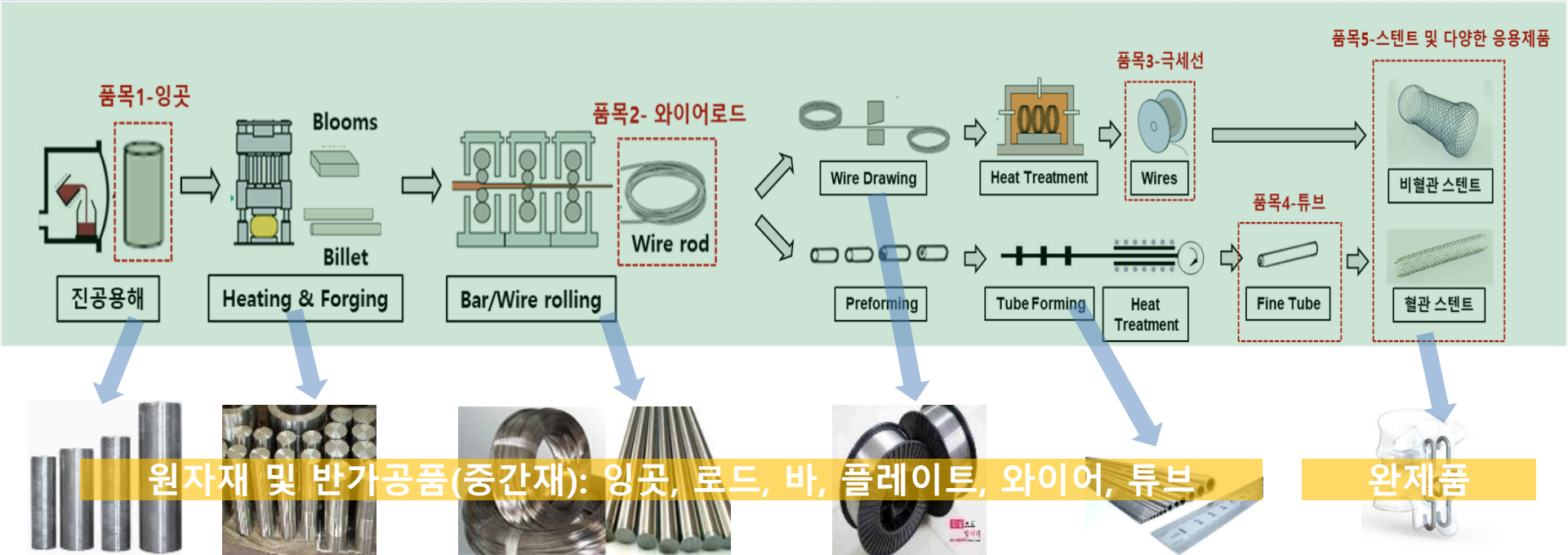
2. 기술 및 제품

3. 판매 제품

Materials
BioMed
tiniko

㈜티니코 주요 기술 1

합금설계 및 용해 기술에서 완제품까지 생체의료용 TiNi 초탄성 소재 제조용 전 공정 노하우 확보



공정 예: 시술용 커테터 가이드와이어(Guide wires), 스텐트용 와이어(Stent wires)

(표면처리 공정에 따라 와이어의 색이 변함.)



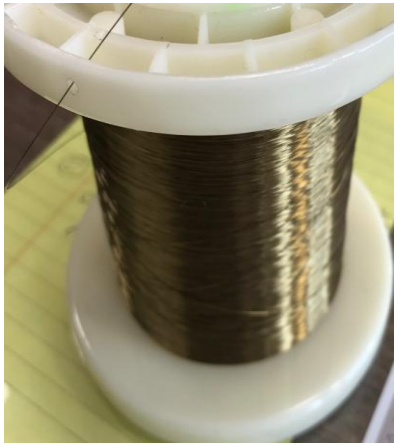
Light Oxide



Etched



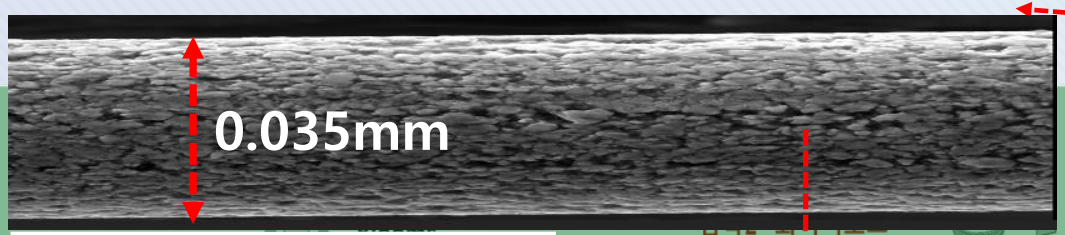
Gloss type

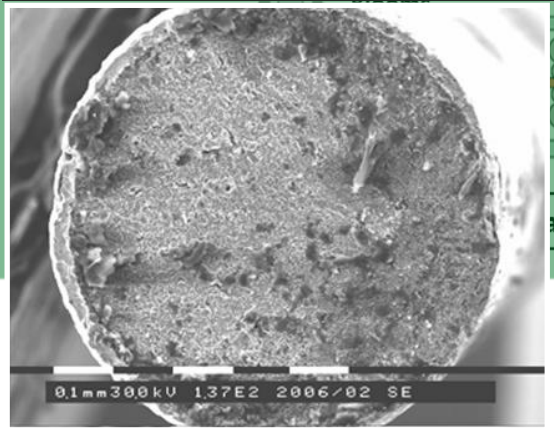


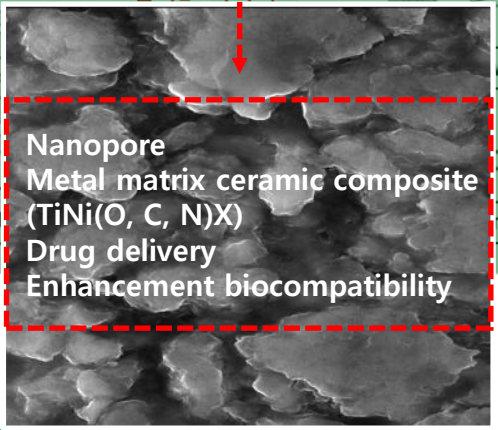
Amber color

㈜티니코 주요 기술 2

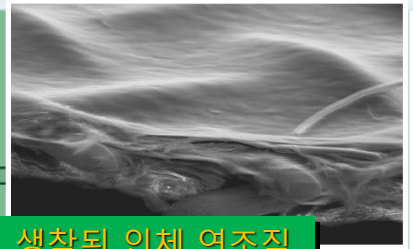
머리카락 굵기의 10배 작은 초탄성 특성의 TiNi 니티놀 극세사 제조 기술

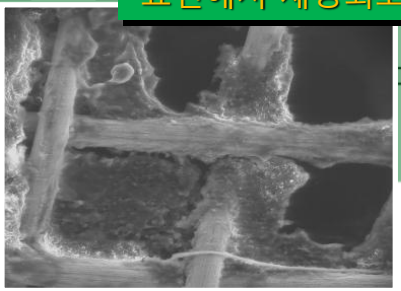


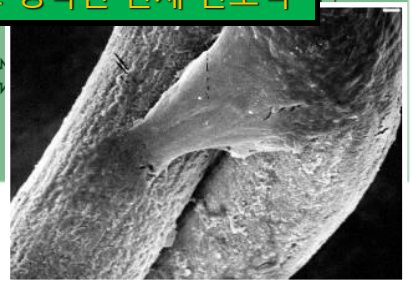












표면에서 재생되고 생착된 인체 연조직

표면 – 나노기공으로 구성된 (flaky surface), 내부 – 고밀도 초탄성

스텐트(Stent) / 인공혈관(Artificial graft)

비혈관용(식도, 담도, 췌장, 등), 혈관 내부/외부용



동물시험



생분해성 인공혈관과 동시 사용



임상시험

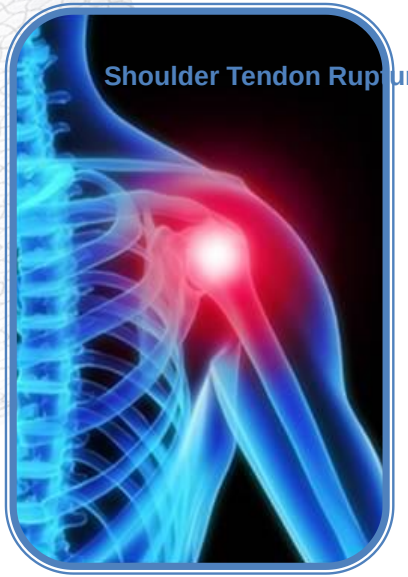


방사선사진

인공인대, 건, 생체뼈, 등 인체 경조직 및 연조직 대체물



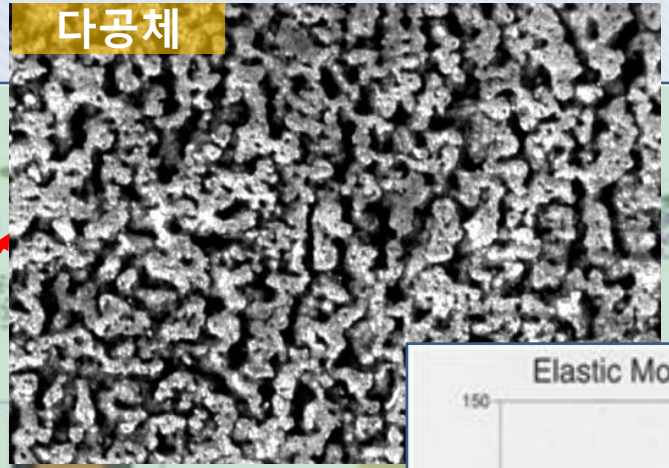




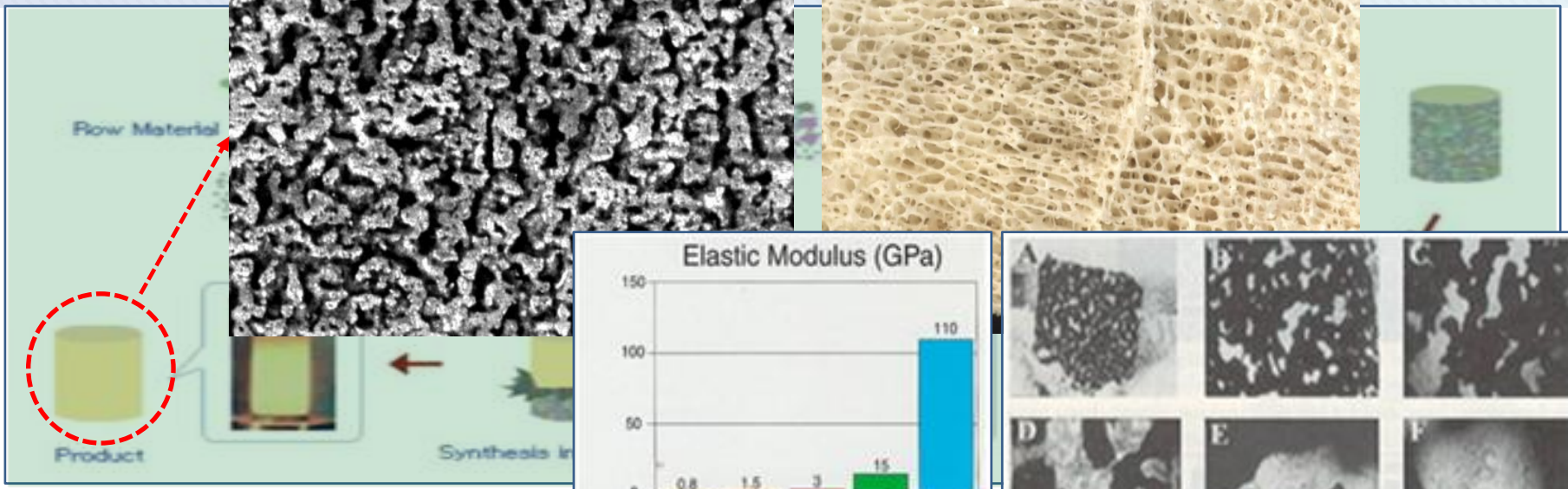
Next generation of superleastic (SE) KNM showing unique performance – a new milestone of surgical management

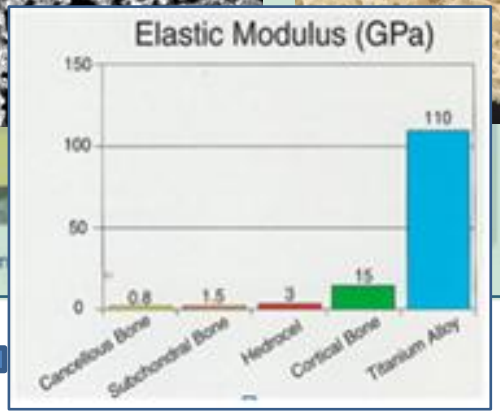
㈜티니코 주요 기술 3

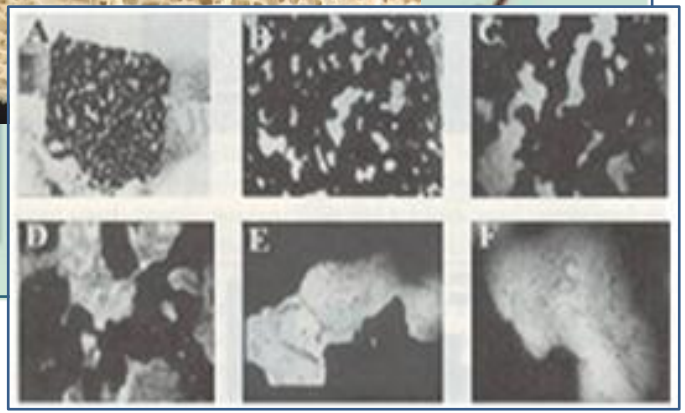
인체뼈와 동일한 기공구조 및 역학적 특성의 TiNi 다공체 제조기술





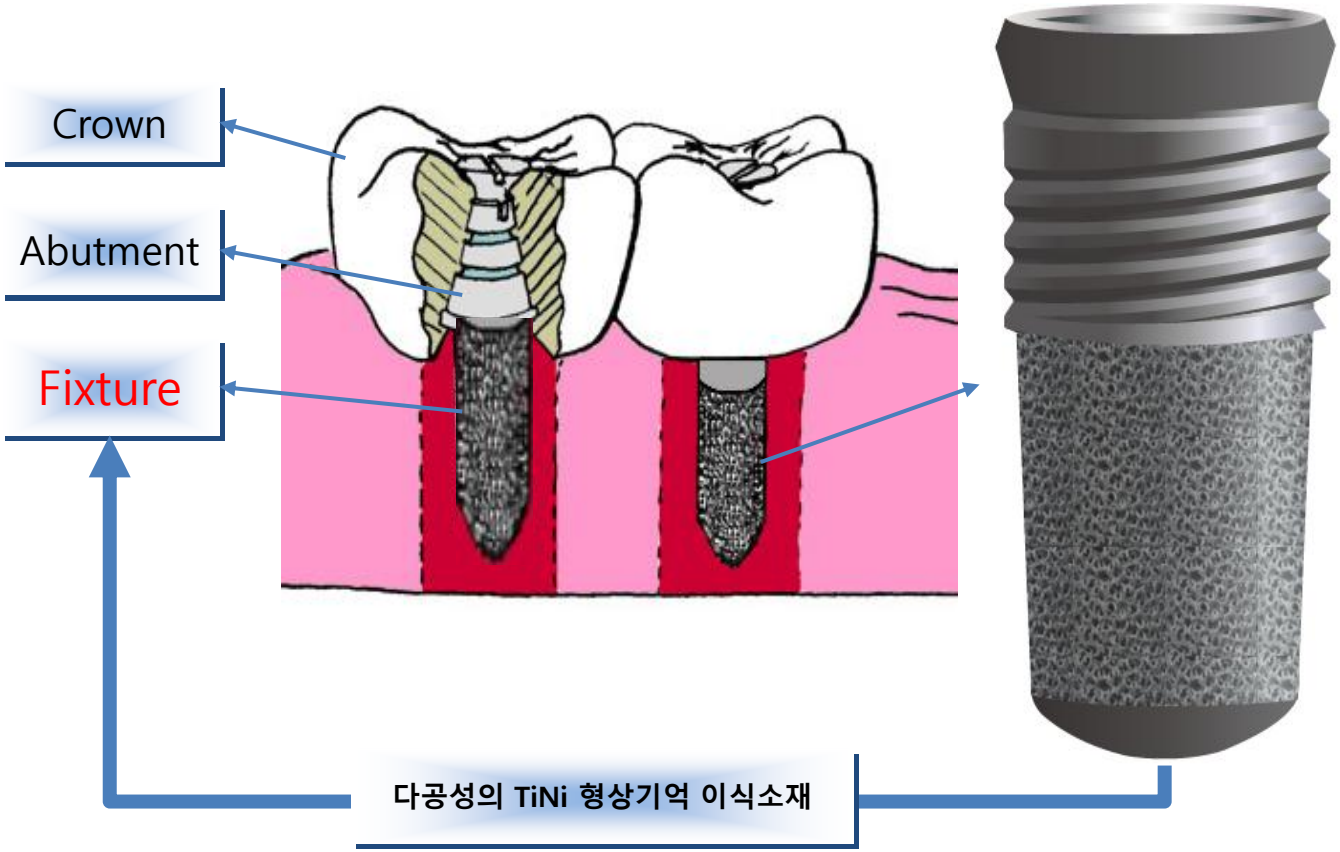






3D Printer 용 powder

다공체 소재 응용 예 1 : 치과용 임플란트 (fixture 파트)



다공체 소재 응용 예 2 : 약물전달기

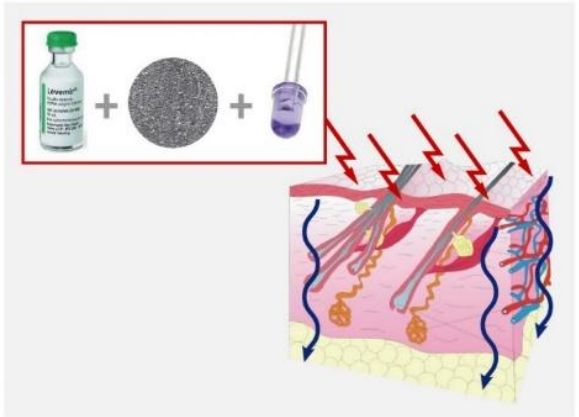


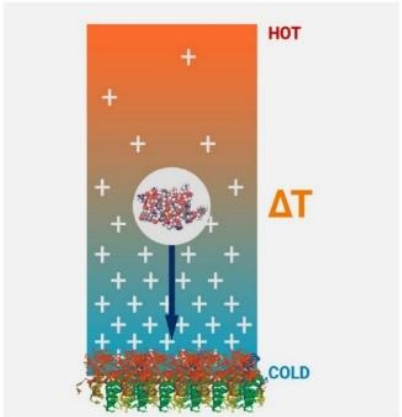




Method of Needle-Free Administration of Insulin

The essence of the NFD (NoNNoS) is to convert electrical energy into the heat flow by means of the IR providing the thermal effect (THERMOPHORESIS – driving force of the NFD).





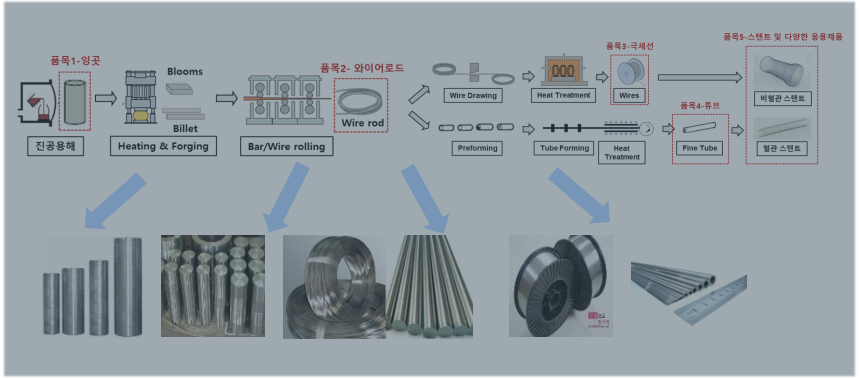
KANG&PARKMEDICAL

- 목 차 –

1. 생체의료용 초탄성 니티놀 소재
(=Nitinol, =TiNi superelastic biomaterial)

2. 기술 및 제품

3. 판매 제품



Materials
BioMed
tiniko

척추용 제품: SpineLant ™

tiniko

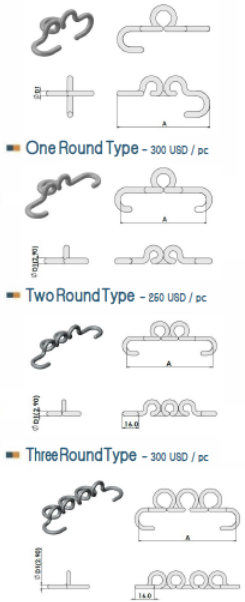
SPINElant™

Normal Type – 250 USD / pc

One Round Type – 300 USD / pc

Two Round Type – 250 USD / pc

Three Round Type – 300 USD / pc



Tolerance:±5%

No.	Model name	D1, Ø	A, mm
Normal Type			
1	SPINElant CESL 1621	1.6	21.8
2	SPINElant CESL 1826	1.8	26.4
3	SPINElant CESL 1831	1.8	31.4
4	SPINElant CESL 2036	2.0	36.0
5	SPINElant CESL 2240	2.2	40.6
One Round Type			
6	SPINElant LESL 2924	2.9	24.2
7	SPINElant LESL 2929	2.9	29.2
8	SPINElant LESL 2934	2.9	34.2
9	SPINElant LESL 2939	2.9	39.3
10	SPINElant LESL 2944	2.9	44.2
11	SPINElant LESL 2949	2.9	49.2
12	SPINElant LESL 2954	2.9	54.2
13	SPINElant LESL 2959	2.9	59.2
Two Round Type			
14	SPINElant LESL 2964	2.9	64.2
15	SPINElant LESL 2969	2.9	69.2
16	SPINElant LESL 2974	2.9	74.2
Three Round Type			
17	SPINElant LESL 2979	2.9	79.2
18	SPINElant LESL 2984	2.9	84.2
19	SPINElant LESL 3289	3.2	89.2
20	SPINElant LESL 3294	3.2	94.2
21	SPINElant LESL 3299	3.2	99.2

Material

SPINElant is made of superelastic TiNi SMA showing shape memory effect (also known as Nitinol which has been already standardized on the ASTM F2063 as the biocompatible material for medical devices and surgical implants). It characterized by high mechanical performance, high cycle fatigue (over 10^7), and improved corrosion resistance.

Features

SPINElant follows the concept of dynamic spinal loops, which is currently gaining increasing recognition

Indications

- P.L.I.F. and posterior fixation
- After decompressive laminectomy
- Correction of degenerative kyphoscoliosis
- Mild degenerative spondylolisthesis
- Compression fracture with kyphosis
- Prevention to adjacent segmental instability
- Injury of ligamentous structure
- Resurgery of F.B.S.S.

Benefits

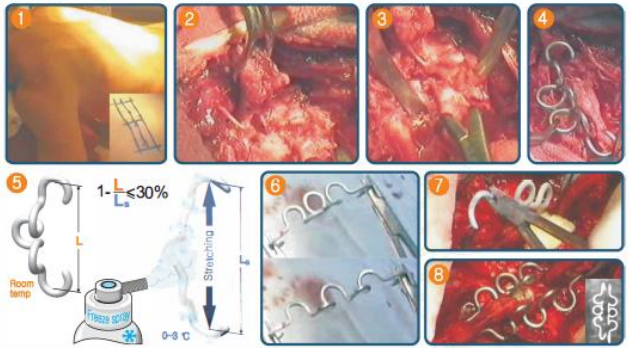
- Flexible, but strong posterior tension band
- Combination with pedicle screw, as appropriate
- Easy to apply and remove
- Correction of kyphosis
- Stable adjacent segment
- Less invasive approach
- Shortening operation time
- Early patient's recovery

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SPINElant™

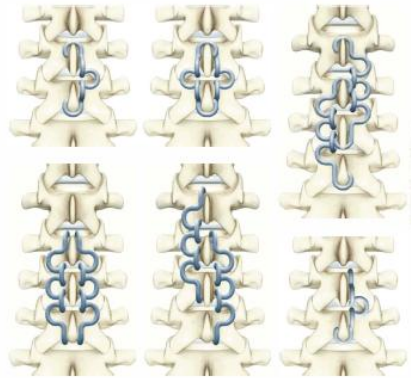
Surgical technique



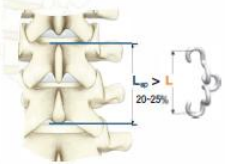
(1) Pronation and demarcation
(2,3) Revision/checking steps (decompression, laminectomy, etc.)
(4) Trying in situ
(5) Cooling (spray or ice water)
(6) Stretching
(7) Insertion
(8) Contraction and X-ray check

Note that the stretching strain should not exceed 30 % of the reference length

Variant embodiments



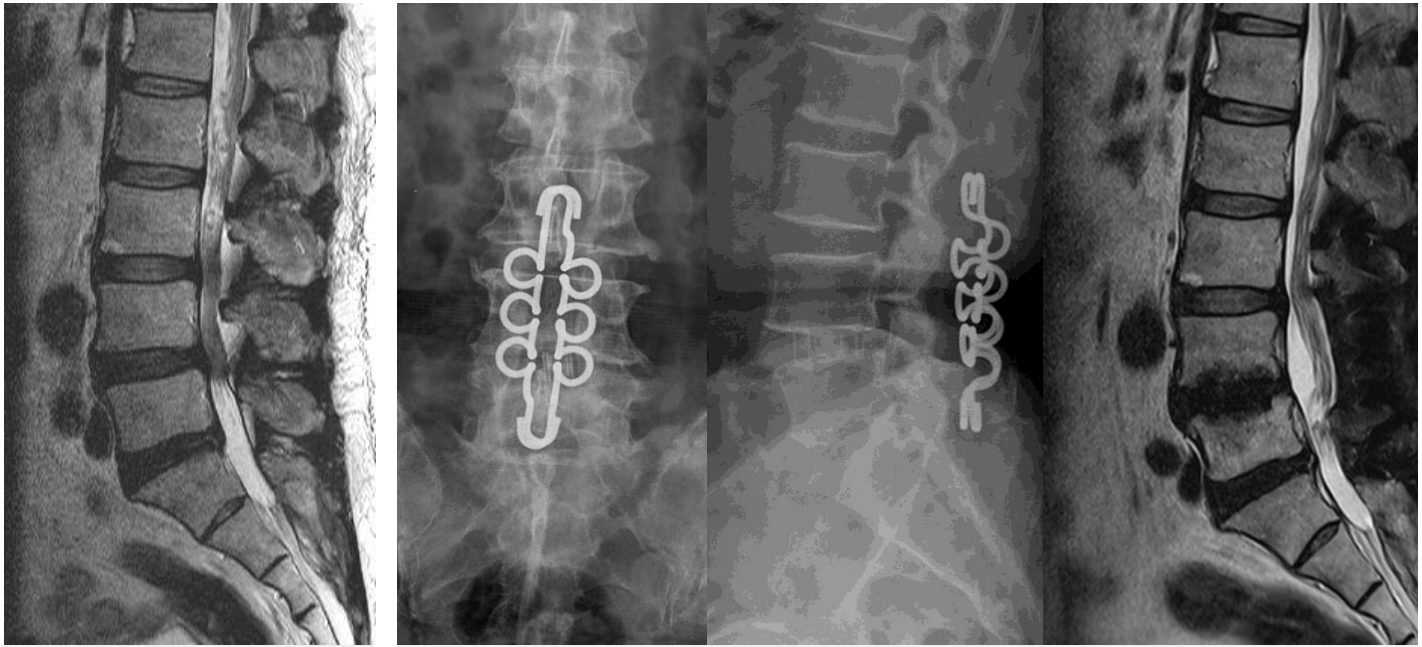
Note that the lenght (L) of SPINElant is to be 1/4 to 1/5 less than that after contracting, complying with the gap (L_{Ss}) between the spinous processes (laminae) to be secured



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연세대, 한림대, 서울대, 등 병원 납품과 해외진출 (인도, 유럽, 등)

형상기억 특성으로 최소침습, 초탄성 특성으로 치료한다.



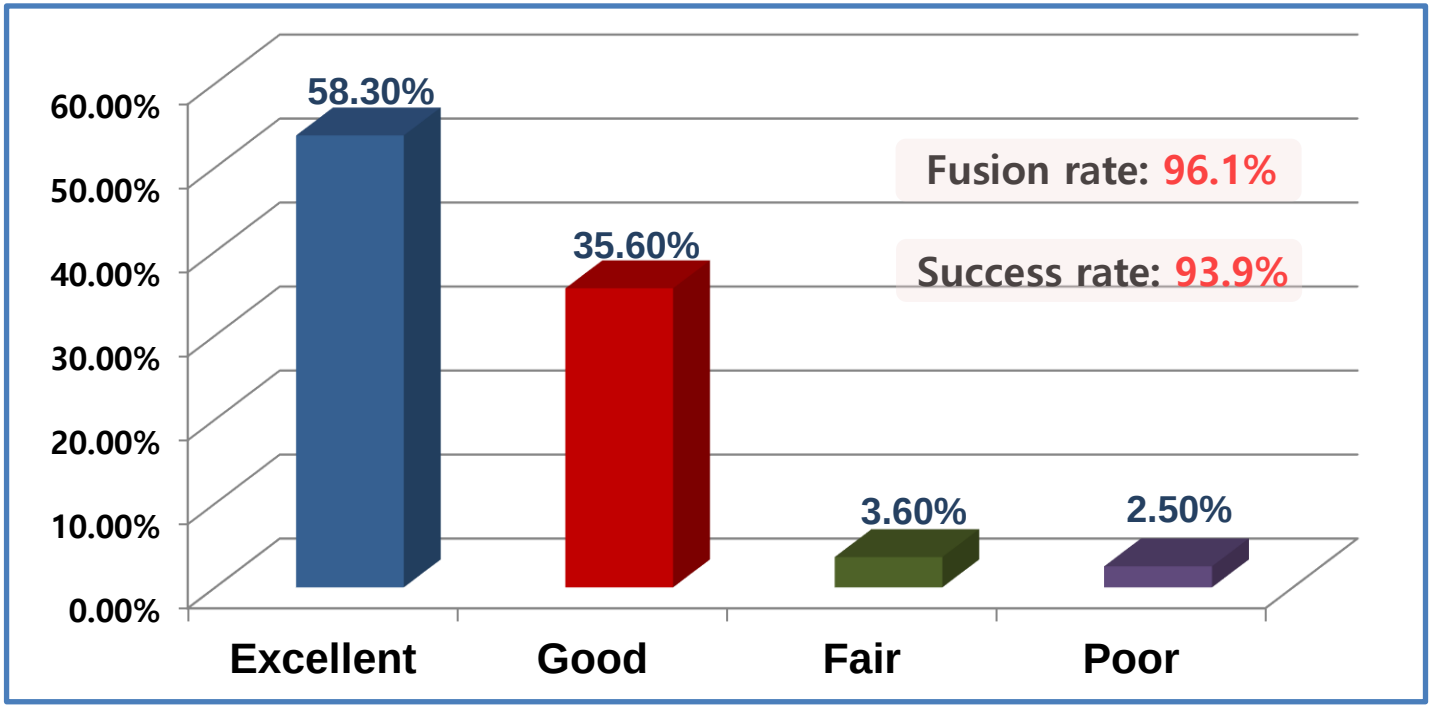
수술 전

수술 후, 5년

Clinical results(Prolo's scale)

ASD Problem 96% 제거 효과
(세브란스, 토마스병원, 보라매병원, 등)

(n=1,454)



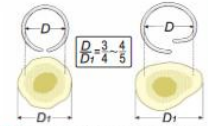
골절 치료 제품 : BoneLant ™

tiniko

BONElant™

Model – 260 USD / pc

Model Type	Round (T-2.A)	G-Shaped (T-2.B)	D-shaped (T-2.C)	D-flattened (T-2.D)	Delta-shaped (T-2.E)	Ellipse-shaped (T-2.F)
Double-ring						
Single-ring						



Note that the optimal compression force requires the inner diameter (D) of the TiNi SMA item be 1/4 ~1/5 less than the outer bone diameter (D1), whereas the shape should resemble the repaired bone circumference taken from MRI pattern

Sizes

Item#	Diameter(D), mm										
	7	10	12	15	17	20	22	25	27	30	35	40
T-2.A	◔	◔	◔	◔	◔	◔	◔	◔	◔	◔	X	X
T-2.B	X	◔	◔	◔	◔	◔	X	X	◔	◔	◔	◔
T-2.C	X	◔	◔	◔	◔	◔	◔	◔	◔	◔	X	X
T-2.D	X	◔	◔	◔	◔	◔	◔	◔	X	X	X	X
T-2.E	X	X	X	X	X	X	◔	◔	◔	X	X	X
T-2.F	X	X	X	X	X	X	X	◔	X	◔	◔	◔

Variant embodiments



Tibial/Fibial fractures

Item#	D, mm
T-2.A	25, 27, 30
T-2.B	27, 30, 35
T-2.D	20, 22, 25
T-2.E	22, 25, 27



Humeral fractures

Item#	D, mm
T-2.A	17, 20, 22
T-2.B	15, 17, 20
T-2.C	15, 17, 20
T-2.D	20, 22, 25



Antebrachial fractures

Item#	D, mm
T-2.A	7, 10, 12
T-2.B	10, 12, 15
T-2.C	12, 15, 17
T-2.D	12, 15, 17



Femoral fractures

Item#	D, mm
T-2.A	22, 25, 27
T-2.B	30, 25, 40
T-2.C	25, 27, 30
T-2.F	25, 30, 35, 40



Clavicular fractures

Item#	D, mm
T-2.A	7, 10, 12
T-2.B	10, 12, 15
T-2.C	10, 12, 15
T-2.D	12, 15

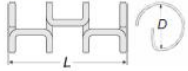
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BONElant

Complicated fractures Type – 350 USD / pc

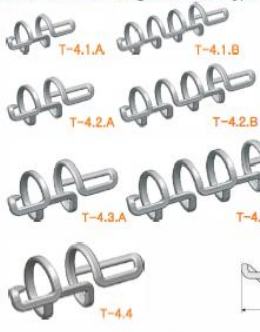




Item#	D, mm	L, mm
T-3.1	12	30
T-3.2	15	34
T-3.3	17	37



Comminuted wedge fractures Type – 400 USD / pc





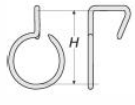


Item#	D, mm	L, mm
T-4.1.A	12	30
T-4.1.B	12	53
T-4.2.A	15	32
T-4.2.B	15	55
T-4.3.A	20	35
T-4.3.B	20	58
T-4.4	25	37



Coracoclavicular fracture Type – 260 USD / pc







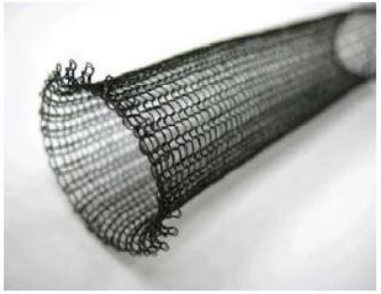


Item#	H, mm
T-5.1.A	26
T-5.1.B	
T-5.2.A	30
T-5.2.B	
T-5.3.A	34
T-5.3.B	
T-5.4.A	38
T-5.4.B	

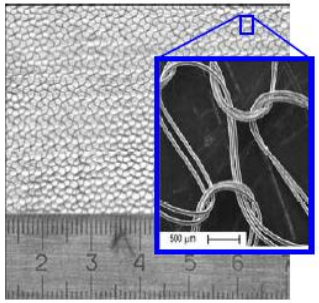


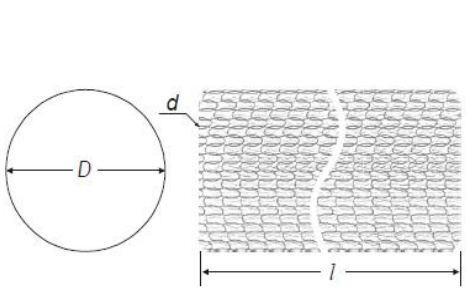
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Mesh 제품 : TiNiKo-M

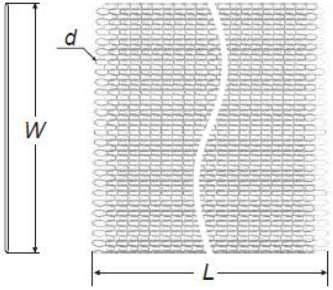




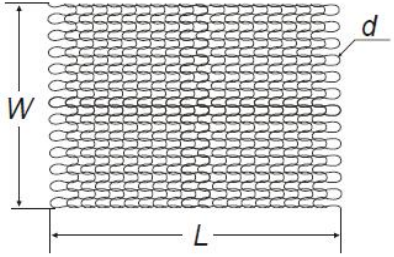




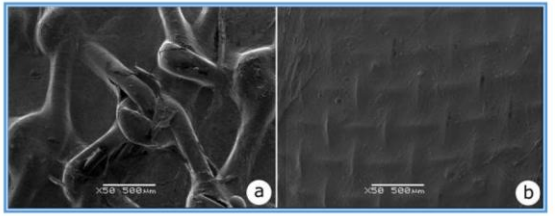
[Cylindrical Sleeve - **CS**]



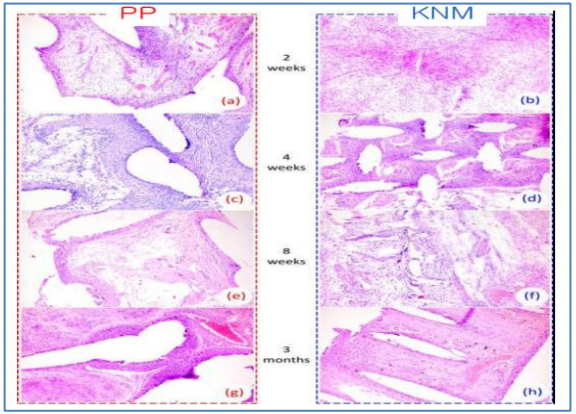
[Flattened Sleeve - **FS**]



[Flat Mesh - **FM**]



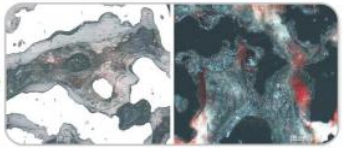
Rat hernia model. SEM image of the incorporated PP mesh (a) and KNM (b) on day 28 post-surgery



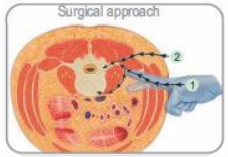
척추 유합용 제품 : SpongyLant ™

SPONGElant™
Porous TiNi TLIF cages

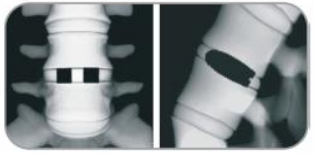
The cages are designed to provide immediate stabilization and earlier fusion via a transforaminal approach using simple and routine surgical procedure. These are made of porous TiNi material showing osteoconductivity, high corrosion resistance and bio-inertness in the body. The distinctive feature of porous TiNi is conditioned by the lowest elastic modulus similar to that shown by the cancellous bone, whereas rheological resemblance in terms of stress-strain allows the cage to be congruentially deformed without rupture and delamination, passing through 10^6 cycles. The rough hydrophilic surface facilitates cell adhesion, growth, and proliferation via a system of interconnected macro-/microvoids and grooves. Rapid bone ingrowth is observed within two weeks, whereas substantial bone through-growth is noted within six weeks. There was no evidence of fibrous connective or fibrocartilaginous tissue formation. Porous TiNi cages sustain superior biomechanical stability and bone-to-graft contact vs. PEEK cages. The implants are recommended to be combined with SPINElant™ system.



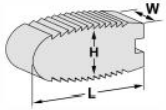
Cortex cage-bone interface Medullary cage-bone interface











(Price 600 USD/pc)

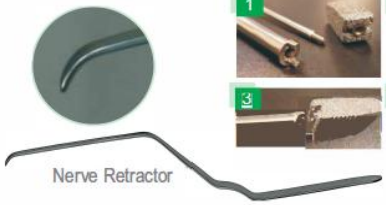
Model name	H-7.1.1	H-7.1.2	H-7.1.3	H-7.2.1	H-7.2.2	H-7.2.3
Length (L, mm)	22	23	24	25	26	26
Height (H, mm)	8	9	10	11	12	14
Width (W, mm)	9	9	9	10	11	11

Porous TiNi TLIF cage system instruments





Trial



Nerve Retractor











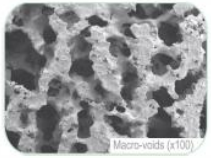
Shape/size can be easily customized

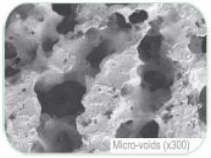


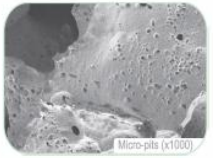


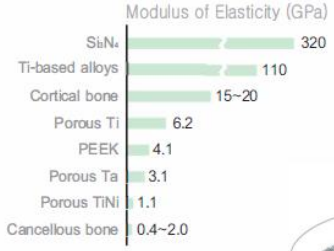












Rival TLIF Solutions



MEDYSSEY



Zimmer Spine





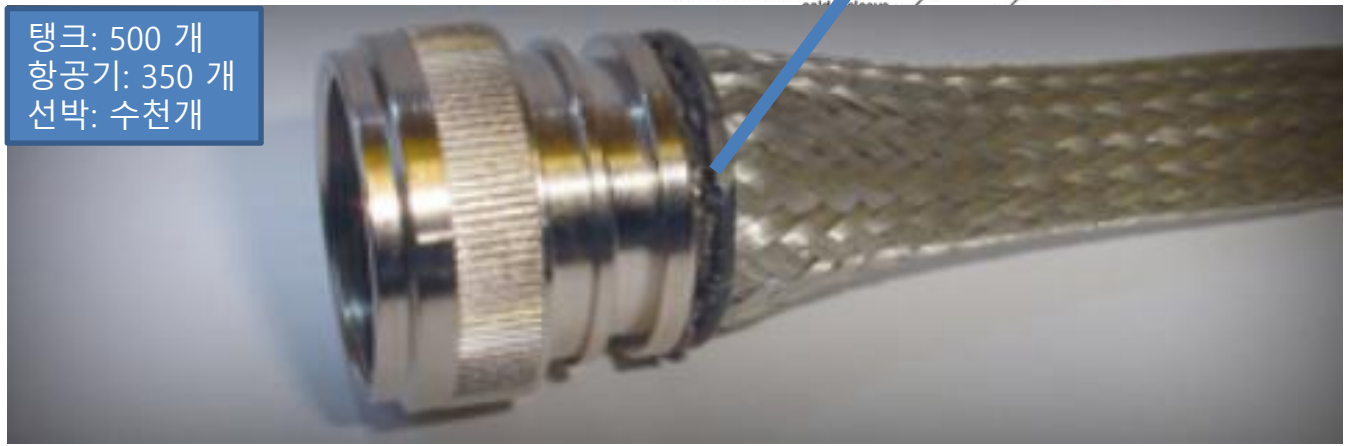
산업용 제품

열수축 링 : 군수용 탱크, 항공기용 전선의 전자파차폐 덮개 조임용





탱크: 500 개
항공기: 350 개
선박: 수천개



천정형 소화기용 스프링클러헤드 제품





제품 요약

매출 품목

구분	품목명
중간재	반가공품 및 열수축링
중간재	장갑용 와이어
중간재	열센서용 엑츄에이터
완제품	척추고정기기 및 외교정 벨트
중간재	스텐트(stent)
중간재	니들(needle)
중간재	수술용 가이드와이어
완제품	추간체유합보형재
완제품	정형외과 임플란트
중간재	치과용 소재
완제품	무침 약물주입기
합계: 11품목	중간재 7품목, 완제품 4품목

중간재 : TiNi, TiNi-Mo, TiNi-Nb, etc





회사 소개

회사 현황



구 분	내 용	구 분	내 용
회사명	주식회사 티니코 (TiNiKo)	사업자등록번호	134-86-76700
대표이사	강지훈	법인등록번호	131411-0258402
설립일자	2011년 05월 01일	업종	의료기기 제조업
자본금	228백만원	매출실적	4,008백만원 (2023′ 기준)
임직원수	16명	주요사업	TiNi기 초탄성 소재 제조
위 치	충북 청주시 흥덕구 오송읍 오송생명12로 163	회사 홈페이지	http://tiniko.co.kr/

회사 연혁

년 월 일		주 요 내 용
2011	03.16	중소기업진흥공단 청년창업사관학교 입교(안산)
	05.01	㈜강앤박메디컬 법인 설립 (법인등록번호: 131411-0258402)
	05.11	안산 세무서 사업자등록 (134-86-76700)
2012	03.16	본사 이전 (안산 중소기업연수원 → 대구TP 벤처공장)
	12.06	공장 등록 (대구광역시관지 연구특구 내, 대구TP 벤처공장)
2014	05.14	본사/연구소 이전(대구TP 벤처공장 → 충북 오송 CV 센터 507호) 공장 이전(대구TP 벤처공장 → 충북 강내면 저산리 저산궁현길 124)
2016	10.16	오창 충북지방중소기업청 T_팩토리 이전 완료
	12.07	T-팩토리 내, 공장등록 완료
2022	03.01	오송 바이오폴리스 산업단지 내로 신축 이전 완료 예정 **(충북 청주시 흥덕구 오송읍 정중리 721번지 – 별첨 지도 참조)**

년 월 일		주 요 내 용
2011	05.01	㈜강앤박메디컬 법인 설립(자본금 5천만원)
	05.23	중소기업진흥공단 지정 멘토-멘티 결연 (멘토사: ㈜메타바이오메드, 멘티사: ㈜강앤박메디컬)
2012	03.16	자본금 증자 (5천 → 1억 3천)
	11.24	중소기업진흥공단 투융자복합금융 성장공유사업자 선정 자본금 증자(1억 3천 → 1억 5천), 3억 전환사채 발행
2014	02.14	㈜메타바이오메드사(코스닥상장사, 봉합사 제조) 투자유치
	03.14	자본금 증자 (1억 5천 → 2억)
2017	06.30	기술보증기금 투자유치 (투자금 10억, 35배수), 상환전환우선주 발행
	06.14	자본금 증자 (2억 → 2억2천8백)
2019	07.20	우즈베키스탄 내 합작법인 설립(Tibtexnika - KNP Medical)
2020	04.24	사명 변경 (㈜강앤박메디컬 → ㈜티니코)

설립 및 이전사항 기준
주요 정영사항 변경 기준

회사 연혁

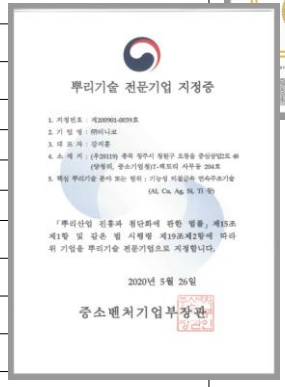
년 월 일		주 요 내 용
2011	10.09	독일 뉘른베르크 국제발명전시회 특별상 수상, 척추고정 임플란트
2012	02.29	중소기업청장 표창 (제9125호), 청년창업사관학교 최우수 졸업
		중소기업진흥공단 이사장 상장 (제2012-003호)
	12.31	중소기업진흥공단 패밀리 기업 지정
		벤처기업확인 (제20121214-35325호, 중소기업진흥공단 대출기업)
		청년창업사관학교 투어한마당 감사패 수상 (창업진흥원 원장)
2013	04.25	기업부설연구소 인증 (제2013111570호, 한국산업기술진흥협회)
	06.28	부품소재전문기업 확인 (제10697호, 산업통상자원부)
2014	05.13	벤처기업확인 (제20140400546호, 기술평가대출기업(중진공))
	06.11	한국산업기술진흥협회 정회원 등록 (제20140425호)
2015	01.20	한국산업기술진흥협회(제20121214-35325호, (사)벤처기업협회)
		한국무역협회 정회원 (무역업고유등록번호: 30764561)
	02.04	충북 창조경제혁신센터 업무협약 및 MOU 체결 협약
	03.09	중소기업진흥공단 글로벌사업화 과정 제1기 우수졸업자 선정
		충북지역 중소기업진흥공단 청년창업사관학교 우수졸업자 선정
	12.06	켑스톤 프로젝트 동남권 은행 상장 (고정 기구)
		켑스톤 프로젝트 울산대학교 LINC 사업단 교내 최우수상 수상
	11.28	병역특례업체 지정 (충북지방 병무청)
	12.16	한국 무역의 날, 충청북도지사 표창장 (제1930호)
2016	10.01	두뇌역량우수기업사업자선정 (바이오분야, 2018년 9월 까지)
	05.24	벤처기업인증(기보, 기술평가보증기업, 제20160106160호)
	06.23	소재부품전문기업확인(제14883호, 2019년 6월 22일 까지)
	11.17	충청북도, 충북경제자유구역청 투자 협약
2017	04.18	충북지방중소벤처기업청 유망창업기업자 선정
	04.19	충북지방중소기업청장 표창장(제2017-17호)
	06.02	한국무역협회 회원등록(회원번호: 30764561)
	08.01	TIPS 프로그램 창업기업부분 기업IR 사업발표 분기 우수상
	12.20	기술보증기금 스타벤처기업 선정 (제31호, 2020년 12월 6일)
2019	10.31	소재부품기업인증(산업통상자원부, 제20414호, 2022년 10월 30일)
	04.29	벤처기업확인(기술보증기금, 기술평가보증기업, 제20200105512호)
	05.06	뿌리기술전문기업 선정(중소벤처기업부, 제200901-0039호)
2024	02.19	InnoBiz, MainBiz 기업 선정

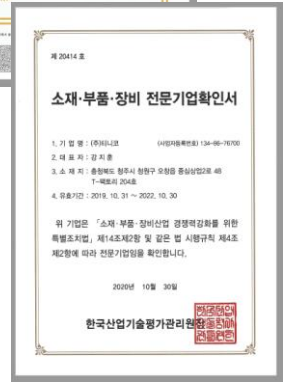
인증 및 수상 기준











회사 연혁

년 월 일		주 요 내 용
2011	09.06	의료기기 판매업 신고 (안산)
2012	04.25	의료기기 제조 및 품질관리 기준 적합인증 (최초 KGMP)
	07.13	의료기기 제조업 허가 (제3809호)
		추간체고정재, EXFLEX 품목허가 (OEM 제품)
		추간체고정재, ANYFIX 품목허가 (OEM 제품)
2013	05.28	의료기기 판매업 등록 (대구, 제637호)
	10.18	추간체고정재 SPINElant™ 품목허가 (제허13-1854호)
2014	09.27	공장 이전 KGMP 인증(KCL-ADB-14002) 및 내삽입용 의료용품)
2015	01.21	추간체고정재 SPINElant-FF™ 수술허가 (제허15-79호)
		의료기기 품질관리 인증 ISO13485 획득
	04.25	추간체고정재 SPINElant-RF™ CE (0120) 인증 (UK, SGS)
		추간체고정재, ANYFIX 품목허가 자진 취하 (OEM 제품)
	12.06	의료기기 판매업 등록 (오송)
		의료용 일반 수술기구 식약처 등록
		추간체고정재 (SPINElant-RF™ 명칭 변경 → INTERfix™)
		추간체고정재, ANYFIX 품목허가 재허가 (OEM 제품)
		허가 품목 한국건강보험심사평가원 코드 등재 신청
2017	03.30	KGMP 이전 신고 완료
		CE 및 ISO13485 이전 신고 완료
	05.01	척추교정벨트 식약처 품목허가 등록완료
		클립 어플라이어 상품 신고 등록
2020	05.20	ISO13485 인증 완료
	06.22	유럽제품인증 완료(척추고정기구 – SPINElant™)
		유럽제품인증 완료(BONElant™)
2024	02.18	무릎근력 강화용 의료기기 Genu Brace 식약처 허가 완료
	02.24	유럽제품인증 진행 중(인공인대)
		유럽제품인증 진행 중(추간체유합보형재, SpongyLant™)
		골절치료기구 식약처 허가 진행 중(BONElant™)

제품인·허가 기준


NOTICE BELGELENDİRME MUAYENE ve DENETİM HİZMETLERİ A.Ş.
Hereby certifies that
TiNiKo Co., Ltd.
With the address(es) below:
Address
has implemented a quality management system according to
EN ISO 13485:2016
The scope of this Quality Management System includes:







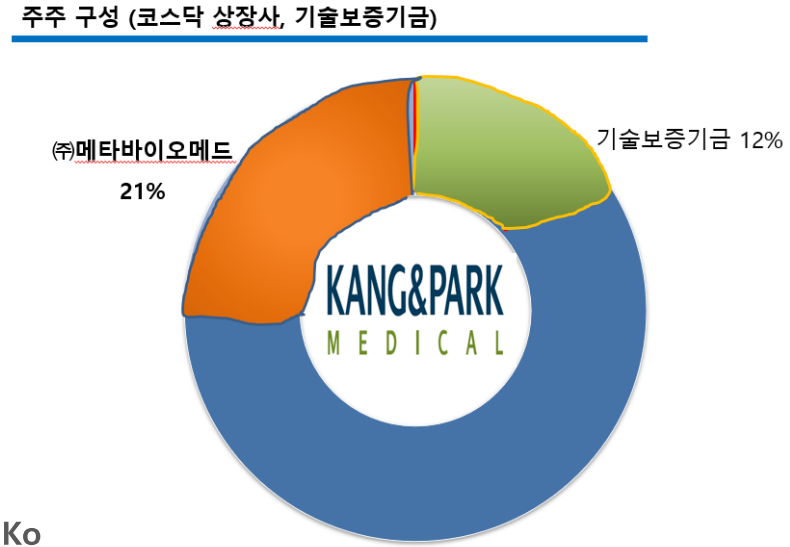
전문성- ㈜티니코

1. 20년 이상 형상기억합금 연구 노하우의 경영진 구성
2. 이노비즈(Innobiz), 메인비즈(MainBiz), 뿌리기술 전문기업(주조, 소성가공)
3. 박사4/석사/1인 연구인력, 매년 1건 이상 SCI 논문 및 특허 출원
4. KGMP, ISO 13485 (ISO 9001), CE인증 보유, US FDA허가 추진 중 (척추용 기구)

KANG&PARK MEDICAL

니티놀 소재 전문기업으로의 이미지 변화 시도

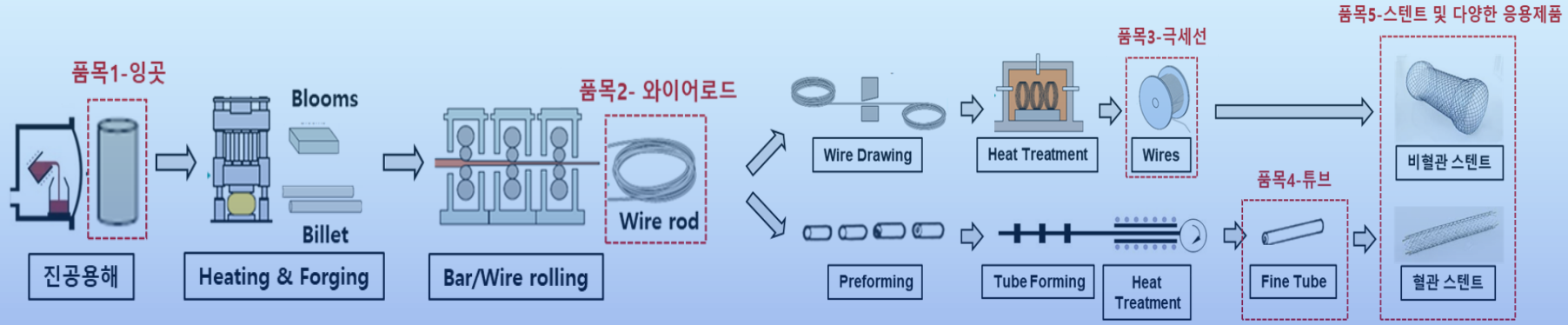
tiniko BioMed



비전 및 정량적 사업화목표



- 2025'내 국내 Top, 글로벌 Top5
- 한국형 SAES Getters 社
- 국민건강 10%(임상효율) 향상
- 수입대체 연 500억
- 내수 1,000억, 수출 1,000억원 달성



감사합니다.

인체조직 특성의 초탄성 TiNi 의료용 소재 국산화기업

All about that, TiNi Shape Memory Biomaterials – TiNiall
TiNi Superelastic Biomaterials of Korea – TiNiKo
주식회사 티니코 (www.tiniko.co.kr)

충청북도 청주시 흥덕구 오송읍 오송생명12로 163
(전화 : 070-7702-8728, 팩스 : 043-237-8728, E-mail : hunywell@gmail.com(기술, 영업), popcorn0930@naver.com(경영), bollessn@naver.com(총무))